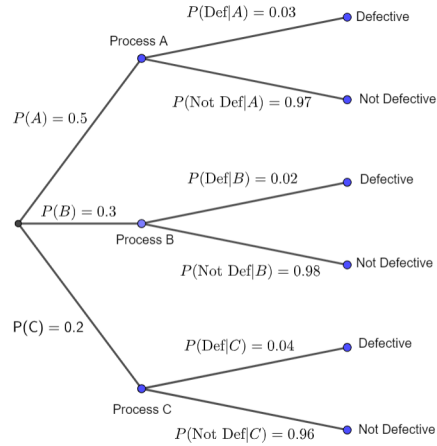


1. Calculating Probabilities using Bayes' Theorem

A computer graphics card is manufactured using three different processes, A , B , and C . The following tree diagram gives the probability that a graphics card is manufactured with each process and the probability of defective and non-defective cards from each process.



Use Bayes' Theorem to calculate the following probabilities. These can be done by hand or in Python.

- If a randomly chosen graphics card is defective, what is the probability it was manufactured using Process A ?
- If a randomly chosen graphics card is not defective, what is the probability it was manufactured using Process C ?

2. Entropy Function for a Probability Distribution

Create a Python function that takes in an array of probabilities $p_1, p_2, \dots, p_n$, from a discrete probability distribution with n possible outcomes and returns the entropy of the corresponding random variable. Use the formula on p. 143 of *Essential Math for Data Science* to calculate the entropy.

3. Calculating Entropies of Probability Distributions

Two random variables X and Y can take on the values $1, 2, \dots, 5$ with the following probabilities.

x	$P(X = x)$	y	$P(Y = y)$
1	0.2	1	0.1
2	0.2	2	0.4
3	0.2	3	0.1
4	0.2	4	0.3
5	0.2	5	0.1

- Use the function you created in the previous problem to calculate the entropies of X and Y .
- Compare the two values found in part (a). Which one is bigger? Explain intuitively why this is the case.